

Catch every strip defect where it is born, not where it is shipped

AI surface inspection that finds, grades and routes every stainless strip defect to the process that caused it.

90%

of defects caught on real mill strip it had never seen

4.7%

of clean strip wrongly flagged, fewer than 1 in 20

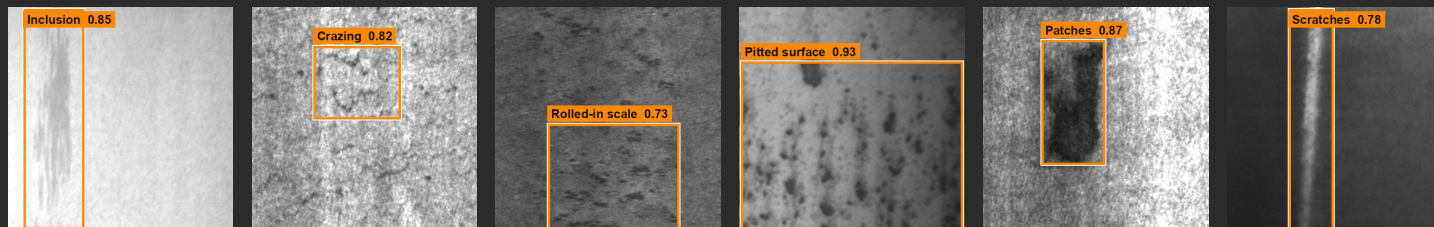
~₹6.7 cr

a year at stake per line, at a 1% downgrade rate

7 months

to shadow mode on one line, go / no-go at month 7

REAL OUTPUT FROM OUR SYSTEM, ON STEEL IMAGES IT HAD NEVER SEEN



TEAM Futuristic | Krishna Kanta Mondal · Prathmesh Walimbe | IIT Bombay



TRY THE LIVE DEMO

surface-vision.github.io

Runs in your browser. Nothing leaves the page.

A defect born at the caster is paid for at every later stage, but today it is found only at the end

Where stainless surface defects are born, and where they are seen today

VALUE ADDED KEEPS RISING →



Melt & cast

SMS

Inclusions, slivers

NOT SEEN



Hot rolling

HSM

Scale, pits, edge cracks

NOT SEEN



Anneal & pickle

HRAP

Pickling patches

SPOT CHECK



Cold rolling

CRM

Scratches, roll marks

SPOT CHECK



Final anneal / BA

CRAP / BA

Stains, BA marks

SAMPLED



Finish & despatch

Slit, cut

Found here, or by the customer

CLAIMS

Why it hurts more on stainless



300 m/min

too fast for the eye to see



₹126k-165k

of nickel in a tonne of 304

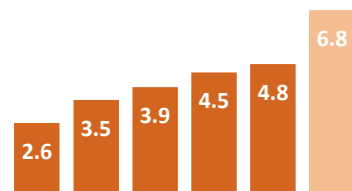


No repair

2B/BA damage can't be polished out

Why now

India demand, Mt **+84% in 5 yrs**



FY21 FY22 FY23 FY24 FY25 FY30F



1.73 Mt imports

FY25; anti-dumping probe on cold-rolled 300/400, Sep 2025



IS 6911 mandatory

BIS certification for sheet and strip, since Aug 2024



2.67 MTPA

JSL cold rolling by FY28, up from 2.05

Cost of quality: where AI moves the money

spend moves from failure to prevention

Prevention

defect sent to its owner in minutes

Appraisal

100% of the surface, every coil

Internal failure

caught before more value is added

External failure

fewer escapes to customers

Poor quality costs ~15% of sales in manufacturing (IISE).

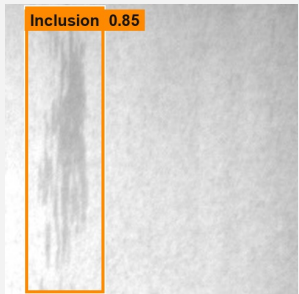
Sources: AMETEK/Ternium; ASSDA; IISE; ASTM A240; LME nickel USD 16,402/t at INR 95.8 (22 Sep 2026); ISSDA; DGTR (Sep 2025); BIS QCO 2024. Photos: Wikimedia Commons (notes).

PROBLEM · WHY NOW

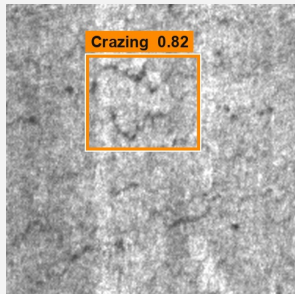
The problem is not that defects exist. They are found late, by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so every defect type must reach the process that caused it

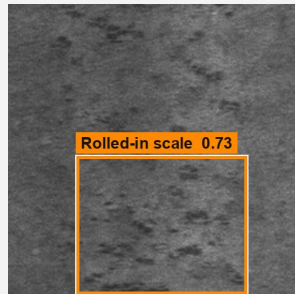
Six defect families, each traced to the process that caused it (real output from our system)



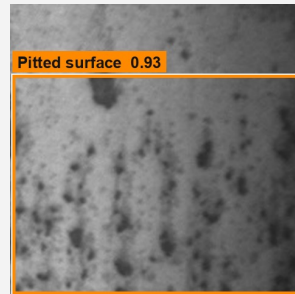
Born: **Caster**
Owner: **SMS**
Fix: Tundish level, mould flux



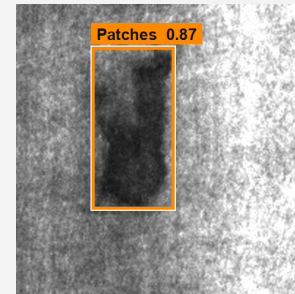
Born: **Reheat, hot rolling**
Owner: **HSM**
Fix: Reheat and coiling temperature



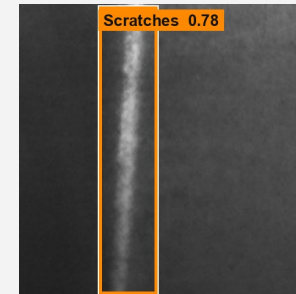
Born: **Hot rolling**
Owner: **HSM**
Fix: Descaler pressure, nozzles



Born: **Reheat, pickling**
Owner: **HSM / pickling**
Fix: Furnace atmosphere, over-pickling



Born: **Anneal & pickle**
Owner: **Pickling line**
Fix: Acid strength, line speed



Born: **Cold rolling, handling**
Owner: **CRM / logistics**
Fix: Guides, rolls, coil handling

Stainless-only types to add with JSL data:

Roping

Ridging (ferritic 430)

Orange peel

Anneal colour

BA stains

Lamination

Precedent: who already does this, and what we copy



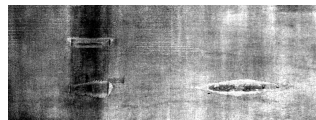
Outokumpu AI inspection live since 2024

Copy: offline, multi-site



Tata Steel video analytics, cold-rolled strip

Copy: in-house team owns it



Parsytec bright + dark-field light on stainless

Copy: optics spec

Grade-by-defect risk: which defects to train for first

Grade	Slivers	Scale	Scratches	Edge cracks	Ridging
200 series (Cr-Mn)	M	M	H	M	L
300 series (304, 316)	H	M	H	L	L
400 series (409, 430)	H	M	M	M	H
Duplex (2205)	M	H	M	H	L

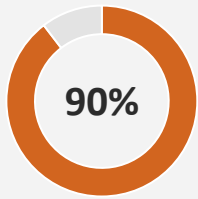


Images: our system on held-out frames. Sources: Nortal/Outokumpu 2024; Tata Steel; Parsytec; Leão 2021; Steel in Translation. Photos: Commons (notes). Owners: team hypothesis. Grade risks: team hypothesis.

Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on 300-series cold-rolled strip.

Four design choices that make the system work on a real line

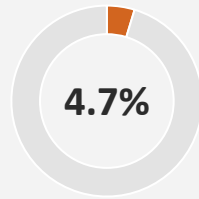
1 Alarms you can act on



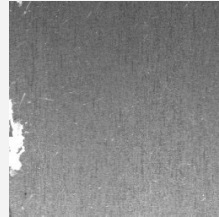
258 of 288 caught



defect: flagged



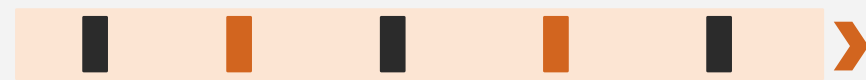
14 of 300 flagged



clean: no alarm

So we: prove it on JSL's own clean and defective strip before any alarm is trusted.

2 One verdict per coil, not a stream of alarms



one coil, both faces: every defect with its type, size and position

Limits per grade and finish: **a light scratch passes on 2B tubes, downgrades a BA panel.**

Our coil report's starting limits (downgrade = in between):

ACCEPT

≤ 2% of frames defective, nothing high or critical

HOLD

≥ 25% defective, any critical frame, or any inclusion



So we: agree accept, downgrade and hold limits per grade and finish with JSL Quality in P0.

3 Image detail sets what is seen, and the hardware bill

0.2 mm per pixel



~18 processors

sees defects from ~0.5 mm

0.5 mm per pixel



~3 processors

sees defects from ~1.3 mm

1.28 mm per pixel



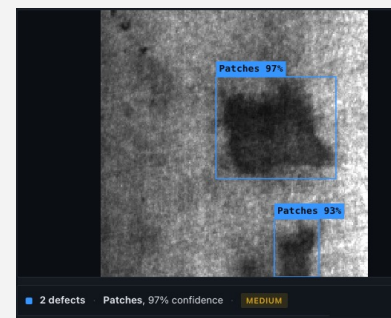
1 processor

fine scratches vanish (~3 mm)

1.28 m strip at 250 m/min. Processors scale with pixels; a defect needs 2-3 pixels.

So we: fix the image resolution with JSL first, then size the hardware.

4 Operators act only on alarms they can trust



Evidence on every alarm



Confidence that means what it says



4 weeks of shadow mode



Every override reviewed

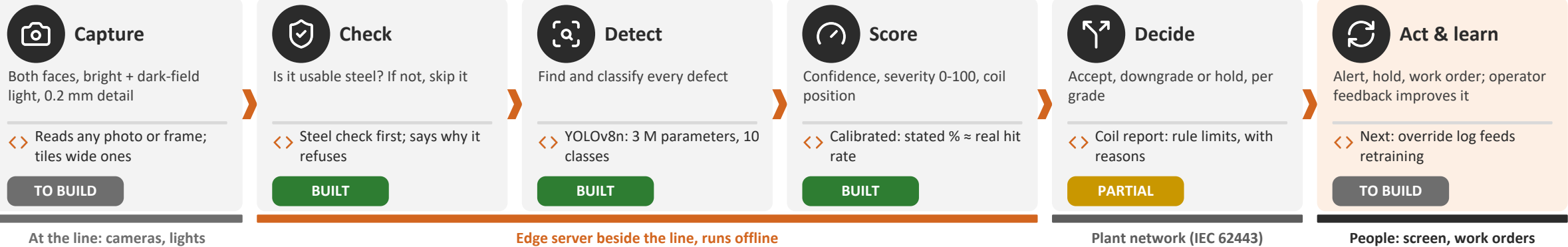
So we: earn trust on the shift floor before the system is allowed to act.

Results measured by us on public steel data the system had never seen. Hardware sizing derived from our line-speed benchmark.

Success is decided on the line, not in the lab: JSL's own strip, the right image detail, coil-level rules and operator trust.

Detect on the strip, decide at the coil, route to the owner, and learn from every correction

How it works: six stages from camera to corrective action, and the software we built for each



Running today: two apps, one model

Browser demo
surface-vision.github.io

Upload a strip image, get:

- box on the defect
- defect type
- confidence
- severity score

16 samples built in, including clean mill strip. Or drop your own photo.

~50 ms an image, nothing uploaded

Operator console
Python app for a line PC

- Steel check on every frame
- Whole-coil batch inspection
- Coil verdict, with reasons
- Line-speed simulation
- Heat map: why the alarm fired

One 12 MB model file runs in both apps. Browser and Python agree on 37 of 37 detections.

What JSL gets that others do not

Alarms you can act on
4.7% false alarms

Root-cause routing
a work order per alarm

One verdict per coil
accept · downgrade · hold

JSL owns the system
no vendor lock-in

Build vs buy

	Cost per line	Proven at speed	Tuned to JSL grades	JSL owns the system
Commercial proven; the vendor owns it	●	●	●	●
Fully in-house JSL owns it; slow to prove	●	●	●	●
Hybrid (recommended) vendor cameras, JSL software	●	●	●	●

● strong
● partial
● weak

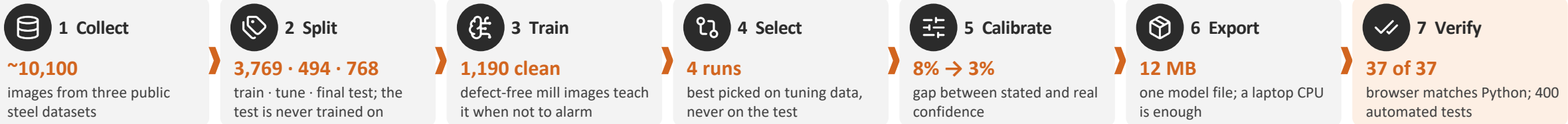
Output: hybrid is never weak.

BUILT = working in our software today. Stack: Python, PyTorch and Ultralytics YOLOv8 to train; ONNX Runtime Web in the browser; Streamlit console. Code: github.com/surface-vision.

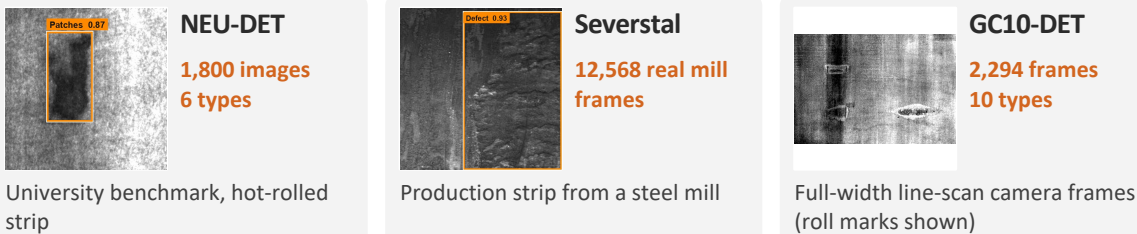
Buy proven cameras, own the software: vendor-grade capture, tuned to JSL's grades and wired to JSL's process owners.

It already works on public steel data; the pilot's job is to prove it on JSL's own strip

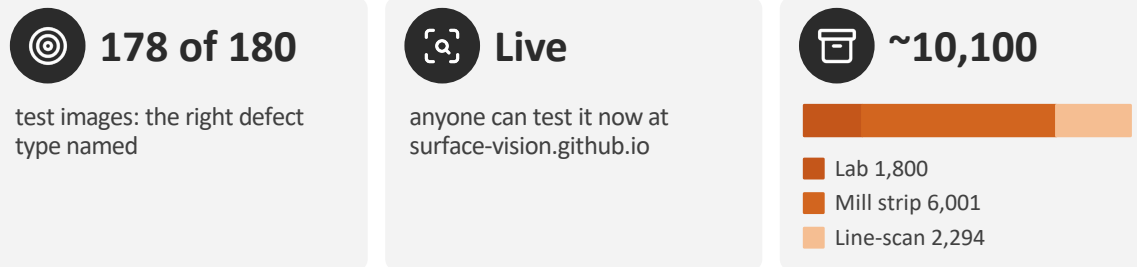
How we built it: seven steps from public images to a model that runs in a browser



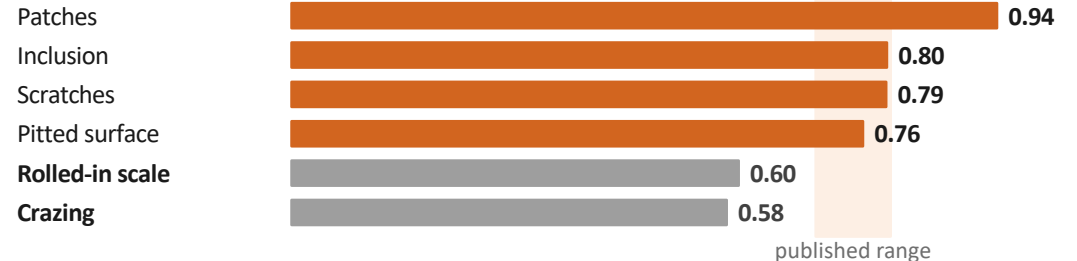
Tested on three public steel datasets



Headline results



How well it finds each defect type (accuracy, held-out images)



Known weak spots, and the fix for each

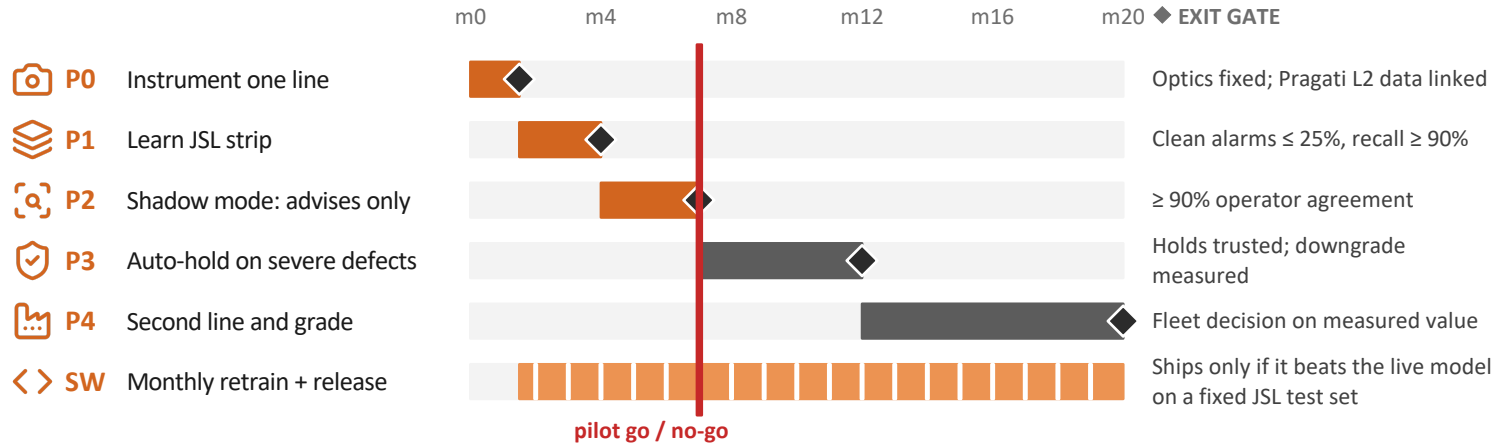
Weak spot	Today (0-1)	Fix in the pilot
Low-contrast texture: crazing, rolled-in scale	0.58-0.60	Angled lighting; JSL labels in P1
Real mill frames are harder than the lab benchmark	0.57 vs 0.75	Prove it on JSL's own strip in P1
Roll marks and edge cracks: few examples	0.20	Edge cameras; targeted labelling with JSL

Datasets: NEU-DET (Northeastern Univ.); Severstal steel defects (Kaggle); GC10-DET (Lv et al. 2020, CC BY 4.0). All scores on held-out images the system had never seen. Build figures: our training and test records.

Strong on four of six defect families today. Every weak spot is known, and each has a fix in the pilot plan.

Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

Stage-gate roadmap (months)



Pilot scope: in and out



IN THE PILOT

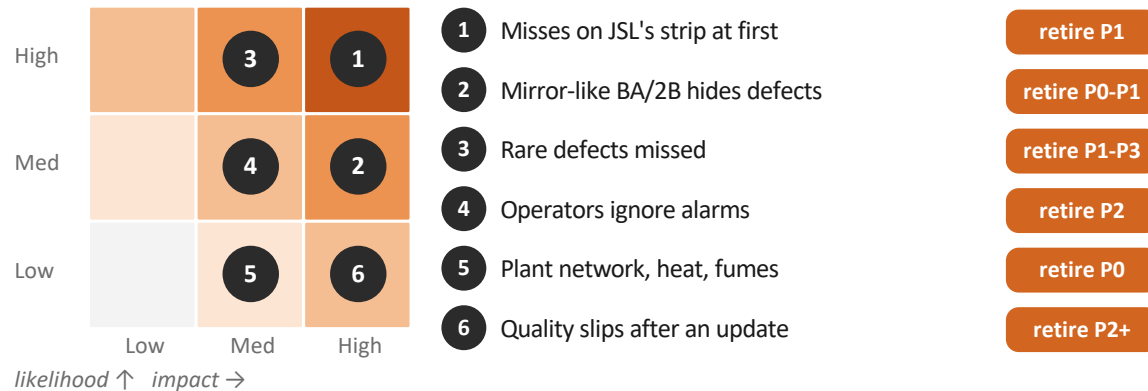
- Both faces, 2B + BA
- 6 defect families
- Read-only until P3

LATER

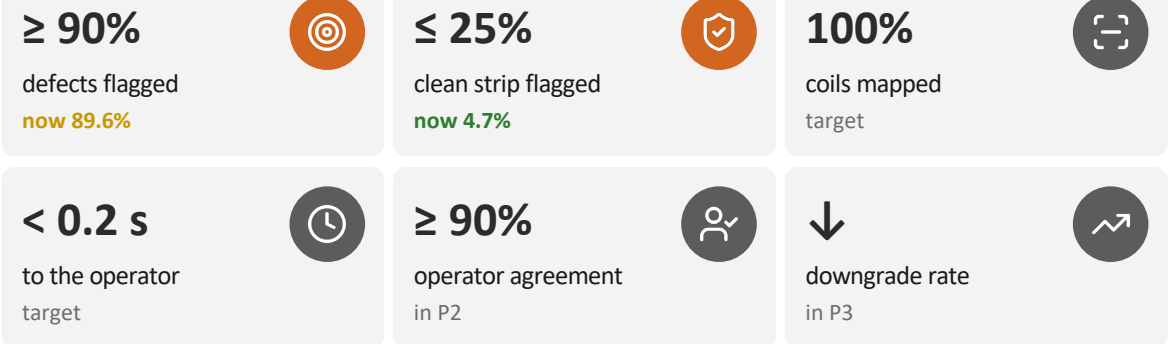
- Hot end, caster side
- Edge cracks
- Other plants

Owner: **JSL Quality** Integration: **Pragati team**

Top risks: where they sit, and when each is retired



Pilot KPIs: how JSL will know it works



Pragati: JSL's digitalisation programme; Phase 2 brings Level-2 process data at Jajpur. KPI targets to be agreed with JSL Quality in P0.

Commit one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

At a conservative 1% downgrade rate, each line has ~₹6.7 cr a year at stake; ~₹43 cr across the fleet

Loss pool per line, ₹ cr a year

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times ₹167,407/\text{t}$

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

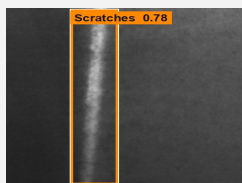
Base case (dark): ₹6.7 cr a year per line, ~₹43 cr across 2.57 Mt. Non-prime coil sells 10-30% below prime.

Where the value comes from, and how the pilot measures it

~₹6.7 cr
a year at stake
per line

- Fewer downgrades** measure: downgrade rate by line and grade
- Fewer customer claims** measure: claims per 1,000 t shipped
- Less rework** measure: rework tonnes per month
- Faster root-cause fixes** measure: hours from defect to fix
- Premium orders** measure: share of exposed-finish orders

Impact beyond the P&L



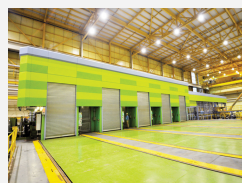
Coverage

every coil, both faces



Customers

a defect map per coil



Process

defect tied to its heat



Safety

hold before a strip break



Sustainability

less rework and remelt



People

inspectors decide, not search

WHAT WE ASK OF JSL

- One 300-series cold-rolled line, 7 months
- Vendor quotes for cameras and lighting in P0
- Downgrade rate by line and grade
- A metallurgist at 20% for root causes

In return: the measured downgrade rate, the value captured, and a go / no-go at month 7.

Inputs: JSL FY26 (INR 167,407/t; 2,565,902 t sold). Base case: a 0.4 Mt line, 1% downgraded at a 10% discount. Value at stake, not a savings forecast: JSL's own data sets the real figure.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. JSL's own downgrade rate decides fleet rollout.